Huaxin Zhang

MSc Artificial Intelligence & Computer Science | Full-Stack Development | Web/Product Prototyping | Research Assistant
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Profile

MSc graduate in **Artificial Intelligence and Computer Science** with experience in **full-stack development, web product prototyping, model training and evaluation, data analysis, technical operations game design and research support**. Worked on **mobility/local-service AI tools, digital museum online exhibitions, EEG/eye-tracking multimodal data mining and depression-recognition** research. Independently completed **prompt-to-play game** generation, browser-extension and emotional-cognitive chatbot projects. Seeking roles in **AI Product Management, frontend/backend/web development**, technical operations, research assistance and AI application-related positions.

Education

University of Birmingham（Dubai Campus）, School of Computer Science | Dubai, UAE | MSc Artificial Intelligence and Computer Science, Merit | 2024.09 - 2025.11

Grades: overall weighted mark 69/100; MSc project (LM Project) 70/100; awarded 30% Merit Scholarship.

Core courses: Machine Learning, Python Programming, Data Structures, Algorithms and Databases, Computer Systems.

Dissertation: Pixel Seed: An AI-Driven Framework for Prompt-to-Pixel World Generation in 2D Games

Dissertation description: This dissertation designed and implemented Pixel Seed, an AI-driven 2D pixel-art web game generation framework. Centred on a "Prompt-to-Play" workflow, the system allows users to generate pixel-style characters, scenes and level content from natural-language prompts and immediately play an interactive demo in the browser. Built with Next.js, React, TypeScript and Zustand, and integrated with the Qwen-Image generation model, it covers the full pipeline from prompt processing and AI asset generation to frontend rendering and gameplay interaction. The dissertation explores AI content generation, web-game architecture, pixel-style consistency, personalised generation and real-time interaction, providing a practical reference for applying generative AI to game-content creation and digital interactive products.

Lanzhou City University, School of Tourism | Lanzhou, China | Bachelor's Degree | 2017.09 - 2021.05

California State Polytechnic University, Pomona | California, USA | Exchange Program | 2018.02 - 2018.07

Core Skills

Product Design: Requirements discovery and analysis, user research, competitive analysis, PRD writing, Axure/Figma prototyping, interaction logic design, MVP and product-roadmap planning	**Data Analysis:** Python data analysis, SQL, user-behaviour analysis, data-metric system design, funnel analysis, A/B-test design and product retrospective
Technical Understanding: Familiar with web frontend logic (Next.js/TypeScript) and backend development logic; understands AI model training, evaluation and deployment workflow, prompt engineering and blockchain fundamentals	**Project Management:** Cross-functional collaboration, development schedule follow-up, requirements review organisation, launch acceptance, grey testing and Agile development process
Industry Experience: AI tool products, Web3 decentralised products, digital cultural-heritage and online exhibition products, short-video content products; practical experience from requirement breakdown and prototype design to launch maintenance and experience review	**Languages:** Chinese native; IELTS 7.0 (Listening/Reading 7.5), able to conduct English product research and overseas-user communication; Japanese conversational

Professional Experience

1. **Yassir** | AI Product Manager | **Abu Dhabi, UAE | 2023.11 - 2024.08**

Worked as an **AI Product Manager** in the mobility and local-services business team, responsible for translating ambiguous pain points from customer-service and operations teams into discussable, schedulable and verifiable product requirements, with focus on **business understanding, solution generation, cross-team alignment, grey validation and post-launch review**.

- **Requirement intake and problem definition:** Collected high-frequency issues from customer-service, operations and user feedback; distinguished process, rule, tool and AI-solvable issues; organised a requirement pool, scenario descriptions and functional hypotheses.
- **Product documentation and interaction expression:** Translated product ideas into PRDs, user flows, low-fidelity prototypes and acceptance criteria, describing feature logic and boundaries in a way that customer-service staff could understand, engineers could break down and operations teams could implement.
- **Collaboration and version management:** Participated in requirement reviews, development scheduling, grey testing and launch acceptance; maintained requirement status, risks and feedback records; promoted small-version iterations through sprint rhythm.
- **Post-launch review and AI capability accumulation:** Reviewed experience issues using feature-usage data and frontline feedback; accumulated prompt templates, knowledge-base rules, standard replies and human-handoff boundaries as a basis for later AI customer-service and operations-tool iterations.

Project Outcome | SmartOps | AI Customer-Service and Operations Collaboration Platform | Product Manager | 2024.02 - 2024.08

- **Methods and outputs:** MoSCoW/RICE prioritisation, Service Blueprint, prompt templates, UAT/grey testing, A/B testing, funnel and path analysis.
- **Scenario research and requirement layering:** Around four high-frequency scenarios -- user travel, delivery orders, driver support and merchant operations -- reviewed tickets, complaints, operations feedback and internal knowledge documents; broke down 180+ valid requirements and defined 8 core MVP features.
- **Service-flow and product design:** Drew user issue-handling paths and a Service Blueprint; clarified the "issue identification - knowledge retrieval - AI-assisted reply - human confirmation - human handoff" chain; designed core modules including an AI conversation workspace, knowledge-base search, order-issue classification and multilingual reply suggestions.
- **Prompt and business-rule implementation:** For order delays, refund enquiries, driver order-taking exceptions and merchant information updates, organised business rules and knowledge-base structures; designed 35+ scenario-based prompt templates and optimised stability through manual review, sample testing and error-attribution analysis.
- **Testing validation and data review:** Used UAT, internal grey release and A/B tests to validate functions, covering 150+ internal test users; drove 3 version iterations and optimised 20+ experience details; AI-assisted reply usage increased by 24%, average handling time decreased by 18%, and internal test satisfaction reached 86%.

2. **Gansu Jiandu Museum** | Digital Exhibition Technical Support | **Gansu, China | 2022.06 - 2023.09**

Participated in the development of the **museum's online platform and digital exhibitions**, focusing on **mini-program content configuration**, online exhibition **frontend maintenance**, exhibition-device integration and **launch testing**. Worked with curation, visual-design and technical teams to transform bamboo-slip cultural relics, exhibition-label text, image materials and audio explanations into digital exhibition content suitable for online communication.

- **Online platform and frontend maintenance:** Maintained mini-program and online exhibition page configuration; handled issues related to artefact detail pages, image viewing, audio playback, link redirection, resource loading and mobile adaptation to ensure stable user-facing presentation.

- **Digital exhibition content launch:** Supported updates for online exhibitions such as "The Era of Bamboo Slips", "Writing on Slips and Silk", "Silk Road Narratives" and "Frontier Life"; checked artefact images/text, audio guides, interactive Q&A and exhibition paths to reduce content errors and experience breakpoints.
- **Cross-team collaboration and acceptance:** Coordinated artefact information, illustration materials, page copy and functional requirements with curation, copywriting, visual and technical teams; participated in pre-launch proofreading, configuration checks, issue recording and regression verification.

Project Outcome | Jiandu Cloud Exhibition | Mini-Program and Online Digital Exhibition Launch Maintenance | Technical Support | 2022 - 2023

- **Methods and outputs:** User Journey Map, Service Blueprint, UAT testing, cross-device compatibility testing and regression testing.
- **Content architecture and exhibition-route organisation:** Based on offline permanent exhibitions and online visiting paths, helped organise the information architecture of "exhibition homepage - theme unit - artefact details - audio explanation - interactive Q&A", improving browsing order and content hierarchy.
- **Digital content configuration and frontend implementation:** Configured and maintained artefact images, titles, exhibition-label descriptions, audio guides, interactive Q&A and page-jump relationships, supporting the launch and update of four sets of online digital exhibitions.
- **Testing validation and operations maintenance:** Participated in pre-launch UAT, cross-device compatibility and regression testing, focusing on page display, text-image consistency, audio playback, button jumps and image loading; recorded and followed up 30+ display, content and interaction issues.

3. Lanzhou University, School of Information Science and Engineering | Research Assistant | Lanzhou, China | **2025.10 - Present**

Supported postgraduate **research collaboration** around **EEG/eye-tracking multimodal** data mining, **depression recognition** and **brain functional network** analysis, including course-case preparation, experimental data processing, model training and evaluation, literature analysis and project documentation.

- **Course teaching and case support:** Assisted in preparing teaching materials, coordinating tutorials and course discussions, reviewing coursework, invigilating exams and supporting course administration; transformed EEG/eye-tracking data analysis and depression-recognition research into teaching cases.
- **Research data processing and experimental support:** Participated in EEG and eye-tracking multimodal research workflows; supported data cleaning, feature organisation, experimental records and result visualisation; used Python, Pandas and Jupyter Notebook for data exploration and staged analysis.
- **Model training and evaluation collaboration:** Under supervisor guidance, participated in depression-recognition model experiments; supported feature selection, tree-model training, parameter tuning and model evaluation; organised results around accuracy, recall and F1-score to compare different feature combinations and model schemes.
- **Literature analysis and research accumulation:** Tracked research related to EEG, eye-tracking, multimodal learning, brain functional networks and mental-health AI; organised literature notes, experimental processes and project documentation to improve research collaboration and teaching transfer efficiency.

Project Outcome | Working Paper: Decentralised Control and Accountability under Regulatory and Adversarial Constraints | Writing in progress | 2025 - 2026

- **Project overview:** Studies how decentralised systems maintain controllability and accountability under regulatory objectives and adversarial behaviour constraints, analysing trade-offs among privacy, decentralisation, enforceability and responsibility tracing.
- **Research framework and problem modelling:** Around the "control-autonomy-accountability" tension in blockchain/decentralised governance, reviewed literature on regulatory constraints, adversarial behaviour, identity responsibility boundaries and incentive mechanisms, and built the research questions and analytical framework.
- **Mechanism comparison and writing progress:** Compared governance mechanisms such as on-chain audit, access control, verifiable execution and reputation/punishment mechanisms; discussed trade-offs among privacy protection, enforceability and accountability, and advanced the paper structure, literature review and case analysis.

Selected Technical Projects

Pixel World | AI-Driven Prompt-to-Playable 2D Game Generation Platform | MSc Project / Product Owner | 2024.03 - 2024.09 | [GitHub]

- **Methods and outputs:** Questionnaire research, in-depth interviews, competitive analysis, user journey map, PRD, Figma prototype, Scrum Sprint, grey testing and metric review.
- **Requirement insight and product positioning:** Through 120 user questionnaires, 22 in-depth interviews and analysis of 3 similar AI-generated game products, clarified the core value of "low-barrier input, instant playability and high creative freedom", and defined the MVP feature boundaries.
- **Product design and iterative implementation:** Delivered a 28-page PRD and Figma high-fidelity prototype covering prompt input, scene generation, core gameplay interaction and work export/sharing modules; used Scrum to iterate over 3 sprints and organised 50 seed users for internal testing, collecting 130+ feedback items.
- **Data review and experience optimisation:** Built a metric system covering generation success rate, average play time, next-day retention and core-feature penetration; used user-path analysis to locate issues in prompt input and generation-result stability; after launch, user generation completion rate increased by 27%.

Web Study Extension | Full-Scenario Browser Learning Assistant Extension | Product Owner | 2025.05 - 2025.09 | [GitHub]

- **Methods and outputs:** User questionnaire, RICE prioritisation model, MVP planning, PRD, high-fidelity prototype, lightweight interaction principles, A/B testing, retention and penetration analysis.
- **User research and requirement planning:** Surveyed 320 students and workplace learners online, identified four core scenarios -- intensive webpage reading, literature translation, focused study and knowledge accumulation -- scored 26 requirements using the RICE model and defined 6 core MVP features.
- **Interaction design and experience refinement:** Designed the extension information architecture and core operation flows, delivering PRD and high-fidelity prototype; for PDF/DOCX local-reading compatibility issues, worked with developers to create a staged adaptation plan.
- **Functional iteration and data review:** Built a user-feedback loop and drove 2 minor version iterations, optimising 15 experience details; designed two A/B plans for the highlight-collection feature, covering 200 active users; the final plan increased feature usage by 22% and first-day feature usage by 30%.

Inner Seed | AI Emotional-Cognitive Companion Chatbot | Product Owner | 2026.03 - Present

- **Methods and outputs:** In-depth interviews, questionnaire research, emotion-label system, dialogue-flow design, prompt engineering, manual annotation, model-effect evaluation, privacy-protection design and prototype testing.
- **User research and scenario breakdown:** Interviewed 30 young users aged 18-28 and combined insights with 200+ online questionnaires to identify three core scenarios -- emotional release, confusion-sharing and self-growth -- and defined the product positioning as "non-medical, light companionship and guidance-oriented".
- **Dialogue system and model tuning:** Built an emotion-label system with 5 major categories and 22 subcategories; worked with the algorithm side to define indicators such as emotion-recognition accuracy, response time and dialogue coherence; designed dialogue flows and prompt templates, completing 3 rounds of model tuning and improving emotion-recognition matching to 82%.
- **Privacy compliance and experience testing:** Designed anonymous chat, local conversation storage and one-click chat-record clearing for high-privacy scenarios; recruited 50 seed users for prototype testing, achieving 85% overall satisfaction and producing follow-up optimisation directions.

Self-Evaluation

With a master's background in **Artificial Intelligence and Computer Science**, I possess a strong combination of **academic research experience**, **web development** skills, and **product implementation** capabilities. I am familiar with **Python**, **front-end frameworks**, and **multimodal data modelling**, and have gained diverse experience through **international education**, **public-sector** projects, **enterprise** practice, and **internet-related** internships. I have strong **communication, coordination, and project management abilities**.

Combining **technical thinking** with a **humanities-oriented** operational perspective, I am skilled at **transforming AI** technologies into **interactive product** prototypes. I have strong **self-learning** ability, **literature research** skills, and **problem-solving** capability, and can **quickly adapt** to roles related to AI products, front-end development, research assistance, and other technology-focused positions. I am **detail-oriented**, rigorous in work, and a **reliable team collaborator.**